

Tutorial - 1

1. Determine whether each of the following is true or false:

- (a) $\emptyset \subseteq \emptyset$
- (b) $\emptyset \in \emptyset$
- (c) $\emptyset \in \{\emptyset\}$
- (d) $\emptyset \subseteq \{\emptyset\}$
- (e) $\{a, b\} \in \{a, b, c, \{a, b\}\}$
- (f) $\{a, b\} \subseteq \{a, b, \{a, b\}\}$
- (g) $\{a, b\} \in 2^{\{a, b, \{a, b\}\}}$
- (h) $\{\{a, b\}\} \subseteq 2^{\{a, b, \{a, b\}\}}$

2. Let $R = \{(a, b), (a, c), (c, d), (a, a), (b, a)\}$. What is $R \circ R$, the composition of R with itself? What is R^{-1} , the inverse of R ? Is R , $R \circ R$, or R^{-1} a function?

3. Let A be a nonempty set and let $R \subseteq A \times A$ be the empty set. Which properties does R have?

- (a) Reflexivity.
- (b) Symmetry.
- (c) Antisymmetry.
- (d) Transitivity.

4. Show that any function from a finite set to itself contains a cycle.

5. Show that in any group of at least two people there are at least two persons that have the same number of acquaintances within the group.

6. Give examples to show that the intersection of two countably infinite sets can be either finite or countably infinite, and that the intersection of two uncountable sets can be finite, countably infinite, or uncountable.

7. Are the following sets closed under the following operations? If not, what are the respective closures?

- (a) The odd integers under multiplication.

(b) The positive integers under division.

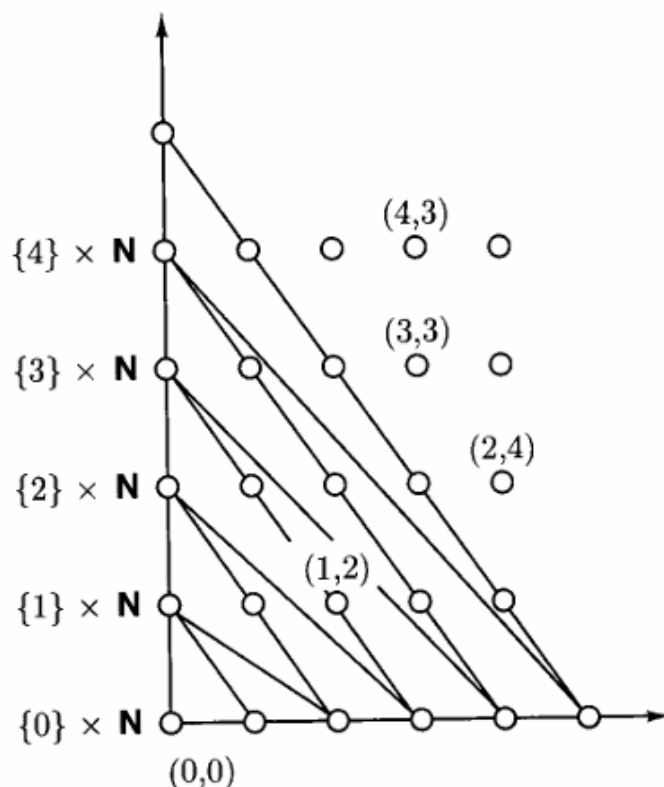
8. What is the reflexive transitive closure R^* of the relation $R = \{(a, b), (a, c), (a, d), (d, c), (d, e)\}$? Draw a directed graph representing R^* .

9. Give an example of a binary relation that is not reflexive but has a transitive closure that is reflexive.

10. The Kleene star of a language L is the closure of L under which relations?

11. Under what circumstances is $L^+ = L^* - \{e\}$?

12. Give the bijection for from $\mathbb{N} \times \mathbb{N}$ to $\mathbb{N}$ for the figure below:



13. Show that, if A and B are any finite sets, then there are $|B|^{|A|}$ functions from A to B .

14. Is the transitive closure of the symmetric closure of a binary relation necessarily reflexive? Prove it or give a counterexample.